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<https://github.com/julianmak/academic-notes>

The repository principally contains the compiled products rather than the source for size reasons.

- ▶ Associated Python code (as Jupyter notebooks mostly) will be held on the same repository. The source data however might be big, so I am going to be naughty and possibly just refer you to where you might get the data if that is the case (e.g. JRA-55 data). I know I should make properly reproducible binders etc., but I didn't...
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OCES 3301 : basic Data Analysis in ocean sciences

Session 2: basic manipulations and statistics

Outline

(Just overview here; for actual content see Jupyter notebooks)

- ▶ basic stats with basic example
- ▶ El Nino 3.4 SST data
 - demonstration of data
 - overview
 - some plotting + exercises

Basic stats

Suppose I have some data samples as the following:

$$x_i = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

- ▶ **sample size** N , the number of samples
- ▶ **range**, largest minus smallest of sample
→ crude measure of spread

averages (but actually three of these):

1. **mode**, most frequent occurrence
2. **median**, rank these, and find the middle one
3. **mean**, THE average

Basic stats

$$x_i = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

- ▶ **lower/upper (25/75 percent) quartile**, rank data, value at which 25/75 percent of data lie below
- ▶ **inter-quartile range**, the different between upper and lower quartile
→ measures spread

Basic stats

Summary as a **box-and-whisker plot**

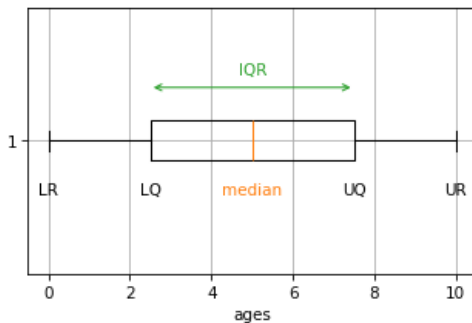


Figure: Nobel prize winning box plot.

Basic stats

$$x_i = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

► **mean**, THE average

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_N}{N} = \frac{1}{N} \sum_{i=1}^N x_i$$

→ sum up, divide by number going into sum

Basic stats

$$x_i = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

► (unadjusted) **variance**

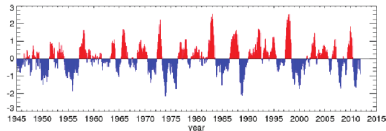
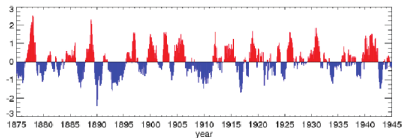
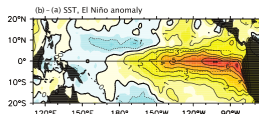
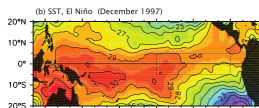
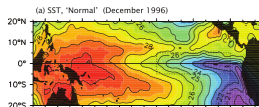
$$s^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_N - \bar{x})^2}{N} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2,$$

→ take mean off sample, square each result, sum, divide by number going into sum

→ square-root of variance is the **standard deviation (s.t.d.)**

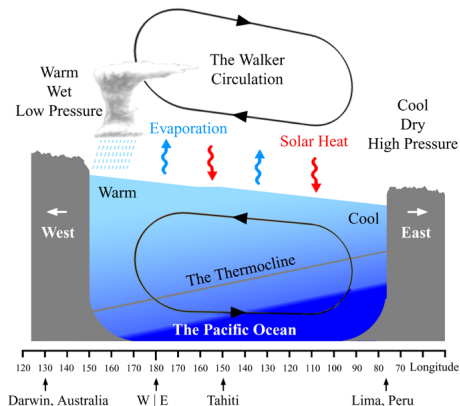
El-Niño (see also ENVS 3004; probably see also OCES 4001)

- ▶ “the little boy”, known to fisherman in South America for a long time
- ▶ generally starts around Christmas time
- ▶ warming in Eastern equatorial Pacific ocean
- ▶ signal in SST in modern day
→ proxy data from corals



Southern Oscillation

- ▶ discovered by Gilbert Walker (1868–1958)
 - correlation with monsoon and thus famine and drought in India
 - Companion of the Order of the Star of India in 1911
- ▶ winds change directions periodically
 - Walker circulation changes (E-W, part of N-S Hadley circulation)



(Ocean heat source moving affects atmospheric circulation)

“Normal” + El-Niño event

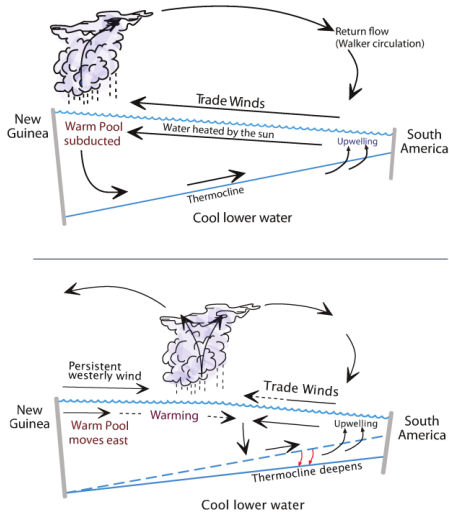


Figure: Schematic of ENSO, from Vallis (2019).

El-Niño 3.4 region

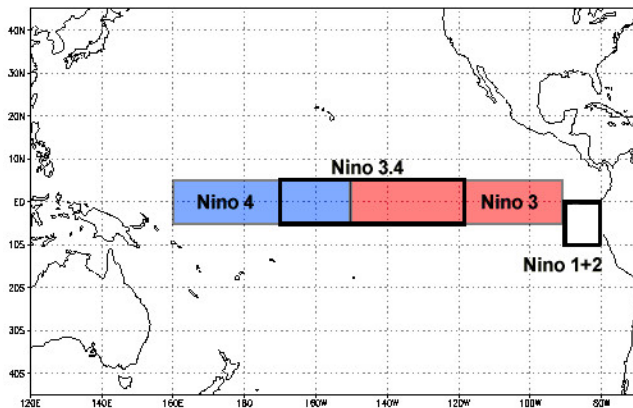


Figure: Pre-defined regions related to El-Niño indices. Picture probably (?) from NOAA.

El-Niño 3.4 SST

Example of time-series data (see S07 and S08 also)

1	1948	1949	1950	1951	1952	1953	1954	1955	1956	1957	1958	1959	1960	1961	1962	1963	1964	1965	1966	1967	1968	1969
2	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99
3	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99	-99.99
4	24.55	25.06	25.87	26.28	26.18	26.46	26.29	25.88	25.74	25.69	25.47	25.29	25.15	25.06	24.97	24.88	24.79	24.70	24.61	24.52	24.43	24.34
5	25.24	25.71	26.90	27.58	27.92	27.73	27.60	27.02	27.23	27.20	27.25	26.91	26.77	26.68	26.59	26.50	26.41	26.32	26.23	26.14	26.05	25.96
6	26.67	26.74	27.17	27.80	27.79	27.18	26.53	26.30	26.36	26.26	25.92	26.21	26.07	25.98	25.89	25.80	25.71	25.62	25.53	25.44	25.35	25.26
7	26.74	27.00	27.57	28.04	28.28	28.12	27.43	26.94	27.01	26.87	26.88	27.00	26.86	26.77	26.68	26.59	26.50	26.41	26.32	26.23	26.14	26.05
8	26.98	27.03	26.90	26.64	27.12	26.80	26.11	25.43	25.12	25.23	25.57	25.26	25.12	25.03	24.94	24.85	24.76	24.67	24.58	24.49	24.40	24.31
9	25.61	25.81	26.22	26.60	26.66	26.55	26.15	25.51	25.28	24.41	24.25	24.57	24.43	24.34	24.25	24.16	24.07	23.98	23.89	23.80	23.71	23.62
10	25.34	25.76	26.46	26.85	27.13	26.81	26.23	25.68	25.73	25.75	25.56	25.71	25.52	25.43	25.34	25.25	25.16	25.07	24.98	24.89	24.80	24.71
11	26.04	26.54	27.46	28.23	28.55	28.36	28.17	27.69	27.44	27.42	27.62	27.90	27.76	27.67	27.58	27.49	27.40	27.31	27.22	27.13	27.04	26.95
12	28.33	28.24	28.27	28.27	28.31	27.99	27.32	26.85	26.40	26.45	26.75	26.62	26.53	26.44	26.35	26.26	26.17	26.08	25.99	25.90	25.81	25.72
13	27.07	27.18	27.47	27.88	27.70	27.37	26.44	26.09	25.92	26.24	26.04	26.18	26.04	25.95	25.86	25.77	25.68	25.59	25.50	25.41	25.32	25.23
14	26.27	26.29	26.98	27.49	27.68	27.24	26.88	26.70	26.44	26.22	26.26	26.22	26.13	26.04	25.95	25.86	25.77	25.68	25.59	25.50	25.41	25.32
15	26.23	26.56	26.94	27.36	27.75	27.67	26.89	26.19	25.78	25.71	26.07	25.97	25.88	25.79	25.70	25.61	25.52	25.43	25.34	25.25	25.16	25.07
16	25.96	26.19	26.80	27.13	27.05	27.08	26.76	26.33	25.94	25.97	25.75	25.67	25.58	25.49	25.40	25.31	25.22	25.13	25.04	24.95	24.86	24.77
17	25.77	26.22	27.18	27.78	27.63	27.62	27.78	27.48	27.40	27.36	27.47	27.62	27.43	27.34	27.25	27.16	27.07	26.98	26.89	26.80	26.71	26.62
18	27.34	27.13	27.02	26.95	26.82	26.59	26.33	25.60	25.32	25.37	25.26	25.23	25.14	25.05	24.96	24.87	24.78	24.69	24.60	24.51	24.42	24.33
19	25.66	26.19	26.94	27.38	27.99	28.09	27.90	27.97	28.01	28.17	28.12	27.96	27.82	27.73	27.64	27.55	27.46	27.37	27.28	27.19	27.10	27.01
20	27.67	27.55	28.21	28.16	27.55	27.64	27.33	26.48	26.27	26.22	26.23	26.03	25.89	25.80	25.71	25.62	25.53	25.44	25.35	25.26	25.17	25.08
21	25.88	26.11	26.50	26.74	27.35	27.47	26.97	26.44	25.86	25.97	26.08	25.95	25.86	25.77	25.68	25.59	25.50	25.41	25.32	25.23	25.14	25.05
22	25.69	25.68	26.33	27.10	27.19	27.88	27.58	27.01	26.72	26.75	27.20	27.27	27.13	27.04	26.95	26.86	26.77	26.68	26.59	26.50	26.41	26.32
23	27.50	27.86	27.82	28.13	28.29	27.69	27.08	27.02	27.15	27.34	27.10	26.98	26.84	26.75	26.66	26.57	26.48	26.39	26.30	26.21	26.12	26.03

Figure: Sample content of elnino34.sst.data.

Palmer Penguin data

Example of categorical + numerical data (multivariate), in `penguins.csv`



Figure: Cursed penguins.

See <https://allisonhorst.github.io/palmerpenguins/articles/intro.html>

Jupyter notebook

Go to 02 Jupyter notebook to play around with these datasets in Python

- ▶ challenge: find from the El-Nino 3.4 data things that are above 90% within some window and call that an 'extreme' event
→ cf. what people do for identifying **marine heatwaves**